Sawyer Rooney

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Mechanical Engineering — Expected 2030

- **Relevant Credit:** 18.01 (Single Variable Calculus), 18.02 (Multivariable Calculus), 8.01 (Classical Mechanics)
- **Current Enrollment:** 3.091 (Introduction to Solid-State Chemistry), 8.02 (Electromagnetism), 18.100B (Real Analysis), 8.UR (Undergraduate Research in Physics)

Southern Maine Community College — Portland, ME
Associate in Science in Computer Science, GPA: 4.0/4.0 — Sept. 2022 – June 2026

- **Relevant Coursework:** Calculus III, Differential Equations + Linear Algebra, Discrete Math, Object-Oriented Design & Programming, Data Structures, Computer Organization, Advanced Artificial Intelligence & Data Science using Python.

Maranacook Community High School — Readfield, ME
High School Diploma, GPA: 4.0/4.0 — Sept. 2022 – June 2026

- **Relevant Coursework:** AP Physics C: Mechanics, AP Physics C: Electricity and Magnetism, AP Chemistry, AP Statistics
- **Selected Honors:** SAT: 1570/1600, Presidential Scholar Nominee '26, Barrowsky Scholar '26, Math Team State Championship '25, Frederick Douglass and Susan B. Anthony Award '25, Rensselaer Medal '25.

Research & Professional Experience

Undergraduate Researcher, Coherent UV Generation via Smith-Purcell Radiation — Oct. 2026 – Present
Massachusetts Institute of Technology — Cambridge, MA

- Processed Lorentzian susceptibility data and fitted Lorentz oscillator parameters using linear regression to model the frequency-dependent dielectric response of silicon in Meep.
- Apply machine learning together with Meep's FDTD electromagnetic simulations to optimize silicon nanowire photonic crystal geometries for coherent ultraviolet Smith-Purcell radiation.

STEM Camp Counselor — May 2026 – Aug. 2026
University of Southern Maine — Gorham, ME

- Developed and led hands-on STEM activities for students ages 7–16, including LEGO-based mechanical demonstrations of gear ratios and introductory lessons on GPU architecture.
- Presented personal projects and programmed simulations in Minecraft to help promote interest in science.

Computer Engineering Intern — June 2025 – Aug. 2025
University of Maine | Maine Space Grant Consortium — Orono, ME

- Integrated a Kinova robotic arm and Intel depth camera using ROS 2 to develop a robotic perception and control system.
- Implemented inverse-kinematics-based motion planning for a Kinova robotic arm using MoveIt.
- Presented an overview of modern inverse kinematics algorithms to University of Maine faculty and graduate students in engineering and computer science.
- Presented research results at a Maine Space Grant Consortium research conference alongside fellow interns and researchers.

Projects

MIT Rocket Team | *Cold-Gas RCS, Feed System Analysis, Additive Manufacturing* — Sept. 2026 – Present

- Designed the CAD geometry of an RCS nozzle in SolidWorks using Rao's thrust-optimized parabolic (TOP) nozzle method, targeting an 80%-length bell nozzle.
- Developed a Python simulation using the Darcy-Weisbach equations to analyze feedline pressure drops and evaluate stagnation pressure available at the nozzle throat.
- Designed testing rigs to characterize SLA-printed photopolymer resin components and experimentally evaluate predicted RCS nozzle performance.

Python | *Machine Learning, Compression, Numerical Methods, Algorithms* — Sept. 2022 – Present

- Developed a Connect Four AI in Python using minimax search with alpha-beta pruning, NumPy, and Pygame.
- Built a handwritten digit classification neural network from scratch using Python and NumPy, implementing backpropagation and cross-entropy loss on the MNIST dataset.
- Developed a Random Forest classifier to predict potentially hazardous asteroid classifications using labels from NASA's NeoWs API, pandas, and scikit-learn.
- Implemented an image compression algorithm using truncated Singular Value Decomposition (SVD), retaining the largest singular values to produce low-rank image approximations in Python with NumPy and OpenCV.
- Implemented QR decomposition and the iterative QR algorithm from scratch to approximate matrix eigenvalues and numerical polynomial roots.